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Information Technology Department

CAPSTONE PROJECT 1 – 2025-2026
Project Proposal and Defense

Date: December 10 - 15, 2025

Purpose: *Ensure that the project is practical, viable for implementation, aligns with the needs and expectations of its intended beneficiaries, and is supported by a well-defined and structured development plan.*

At this stage, students present a detailed proposal of their project. This defense also aims to prepare the students for the upcoming submission of Capstone 1 Progress Report (Chapters 1-3) to be conducted on January 2026 (*exact date will be posted soon*). This defense aims to make clarifications on the project requirements and finalizations on the content of Chapters 1 to 3 of CP Documentation. During the oral defense, the following shall be accomplished by the CP Team:

1. Identify the beneficiary company or organization, including the contact information of its authorized representative (for projects with a sole beneficiary). Proponents may also need to provide proof of the company or organization's legitimacy and existence.
2. Clearly describe the issues or challenges the company or organization faces in its day-to-day operations to establish the relevance and necessity of the proposed project.
3. Define the project's **scope**, the **project's objectives**, its **intended users and stakeholders**, and **relevant use cases**. Specify the **hardware, software, and service requirements**, and outline the **methodologies to be employed for the system's development**.
4. Provide the following diagrams to serve as the foundation for the project's design:
   a. **Flowchart of Existing Processes** – Depicts the current workflows relevant to the system being developed.
   b. **Entity-Relationship Diagram (ERD)** – Illustrates the main entities in the project's data environment, their attributes, and relationships.
   c. **Data Flow Diagram (DFD)** – Maps the flow of data within the system, showing inputs, outputs, and processes.
   d. **System Architecture Design** – Includes user types, network devices, services utilized, and the developed system, offering a comprehensive view of how the system will operate.

5. Thoroughly document the feedback and suggestions provided by the panelists during the defense, and incorporate the necessary changes into Chapters 1-3 of the Capstone Project documentation.
6. Adopt the project title as suggested or approved by the panel, ensuring it aligns with the project's objectives and scope.

Slide Presentation Content:

- **Company/Organization Background and Contact Person-** *Provide a detailed background of the beneficiary company or organization, including the name and role of the contact person. To enhance the presentation, consider including pictures that verify the organization's existence and demonstrate the suitability of the proposed project for its intended recipient.*

  For projects without a sole business/organization beneficiary:
  - *Simply list the primary sources of data, which may include names of individuals or groups who will serve as the primary reference points for the project.*

- **Issues and Challenges Identified** - *Highlight the specific issues and challenges faced by the beneficiaries that are directly relevant to your study. Ensure that these items are clearly*

addressable by the proposed system or computing solution, demonstrating the project's alignment with the identified needs.

- **Main Objective/Purpose of the System –** *Summarize the primary objective or purpose of the system (minimum of 5 sentences). Ensure that the stated objective clearly highlights the system's significance, feasibility, and potential impact, making it compelling for approval as a Capstone Project by the panel.*
- **Scope of the System –** *list here the main modules of the system. System module is component and integral part of the system that performs a specific task.*

  For Example (Enrollment System):

  The modules for an Enrollment System includes:

  1. *User Accounts Management*
  2. *Online Registration for Students*
  3. *Students Online Portal*
  4. *Registration Management*
  5. *Instructors, Subjects and Schedules Management*
  6. *Grades, Student and Enrollment Reports*

  - *<u>Important Note: Include short description for each system module</u>*

- **Users of the System** – *list here the group of individuals who directly interact with the system. This group of individuals are the ones who utilize the system to accomplish a particular task. Please briefly describe here also their specific role in the system/computing solution.*

  For example (Enrollment System):

  System Administrator (School Registrar) - the system admin can view, create, modify, and delete student's registration and academic records. The system admin has also access to all reports that the system may generate and has the capability to manage the users access for the system.

- **Stakeholders –** List all groups or individuals who may be impacted by the development, implementation, or outcomes of the system. This includes not only the system's direct users but also other stakeholders who may benefit from or be affected by the project in various ways. Consider stakeholders such as beneficiaries, decision-makers, external partners, or any other entities that interact with or are influenced by the system's deployment and functionality.

  For example (Enrollment System):

  Parents/Guardians of Students - they are included as one the stakeholders for the system because the school may quickly provide them important school information about their children, which can be obtained from the records stored in the enrollment system.

  Other stakeholders for an Enrolment System may include:

  - *School Registrar Office/Registrar Office Employees*
  - *School Administration*
  - *Students*
  - *Faculty Members*
  - *Developers of the System*

- **Use Cases -** list here the possible situations where the system can be used.

  For example (Enrollment System):

  1. *The enrollment system will be used by students for their 1st, 2nd and short-term registrations in the college which can be done online.*
  2. *The enrollment system can be used to prepare reports for CHED submissions.*
  3. *The enrollment system includes a student portal where student may log in to view their enrollment records and grades.*

- **Hardware, Software and Service Requirements -** List all the hardware, software, and services needed for the system's development, deployment, and user access.

  For example (Web-based Enrollment System):

  1. *Development Requirements*
     - *Code Editor – Sublime Text*
     - *Database – MySQL*
     - *Programming Languages – PHP, HTML, SQL, CSS, Javascript*
     - *Others – Windows 10 OS, Google Chrome, WAMP*
     - *Hardware – Asus Core i5 Laptop*
  2. *Deployment Requirements (On-line)*
     - *Web Hosting Service (Web Server and Software needed to run the system)*
     - *Domain Name Registration Service*
     - *Printers*
  3. *User Access Requirements*
     - *Mobile Device/Laptop/Desktop Computer*
     - *Mobile or Desktop Browsers*
     - *Internet Connection/Mobile Data Service*

- **Methodology -** Present the selected software development methodology, including an illustration to provide a clear understanding of the framework being followed. Discuss each phase of the methodology in detail, ensuring it aligns with the objectives and scope of the project.

- **Flowchart of Existing Processes -** Present a detailed illustration of the existing workflows or processes relevant to the project. The flowchart should visually represent the sequence of activities, decision points, and interactions within the current system or environment.

  - Accompany the illustration with a clear discussion provided by a team member, explaining each step of the flowchart. The discussion should highlight inefficiencies, bottlenecks, or challenges in the existing processes, establishing the basis for the proposed system's development.

- **Entity-Relationship Diagram -** Provide a comprehensive illustration of the Entity-Relationship Diagram (ERD) that represents the project's data environment. The ERD should include all key entities, their attributes, and the relationships between them, clearly defining how data will be structured and managed within the system.

  - Support the illustration with a discussion by a team member, explaining the purpose of each entity, the significance of their attributes, and the nature of their relationships. Emphasize how the ERD aligns with the project's objectives and supports efficient data management and system functionality.

Important Components of an Entity-Relationship Diagram (ERD):

1. **Entities:**
   - Represented in uppercase and in singular form (e.g., STUDENT, COURSE).
2. **Attributes:**
   - Listed inside the attribute box of each entity.
   - Clearly identify all necessary attributes that define the entity.
3. **Relationships:**
   - Typically modeled as one-to-many (1:N) or many-to-many (M:N), with cardinality indicating whether the relationship is mandatory or optional.
   - Include descriptive labels to define the nature of the relationship.
4. **Primary Keys:**
   - Underlined and placed at the top of the attribute box to signify their role as unique identifiers for the entity.
5. **Foreign Keys**:
   - Included in the attribute box of the related entity and marked with the label (FK#) on the left side.

Sample:

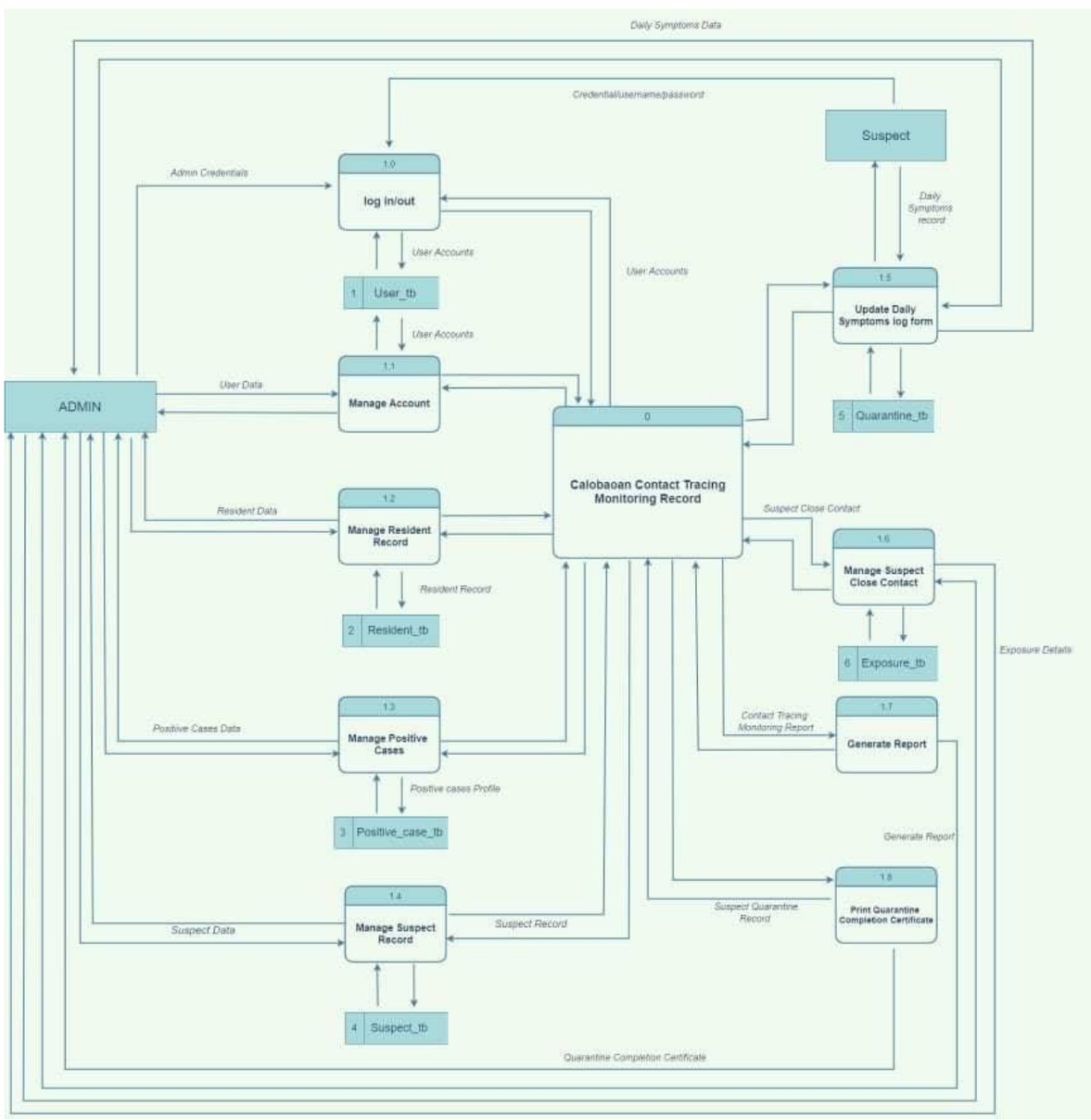
❖ **Data Flow Diagram -** Present a detailed Data Flow Diagram (DFD) that illustrates how data moves through the system. The DFD should include processes, data stores, external entities, and data flows, clearly depicting the interaction and transformation of data within the system.

- Accompany the illustration with a discussion by a team member, explaining each component of the DFD. The discussion should cover the roles of processes, the significance of data stores, and how external entities interact with the system, emphasizing the logical flow of information and its relevance to achieving the project's objectives.

Important Components of DFD:

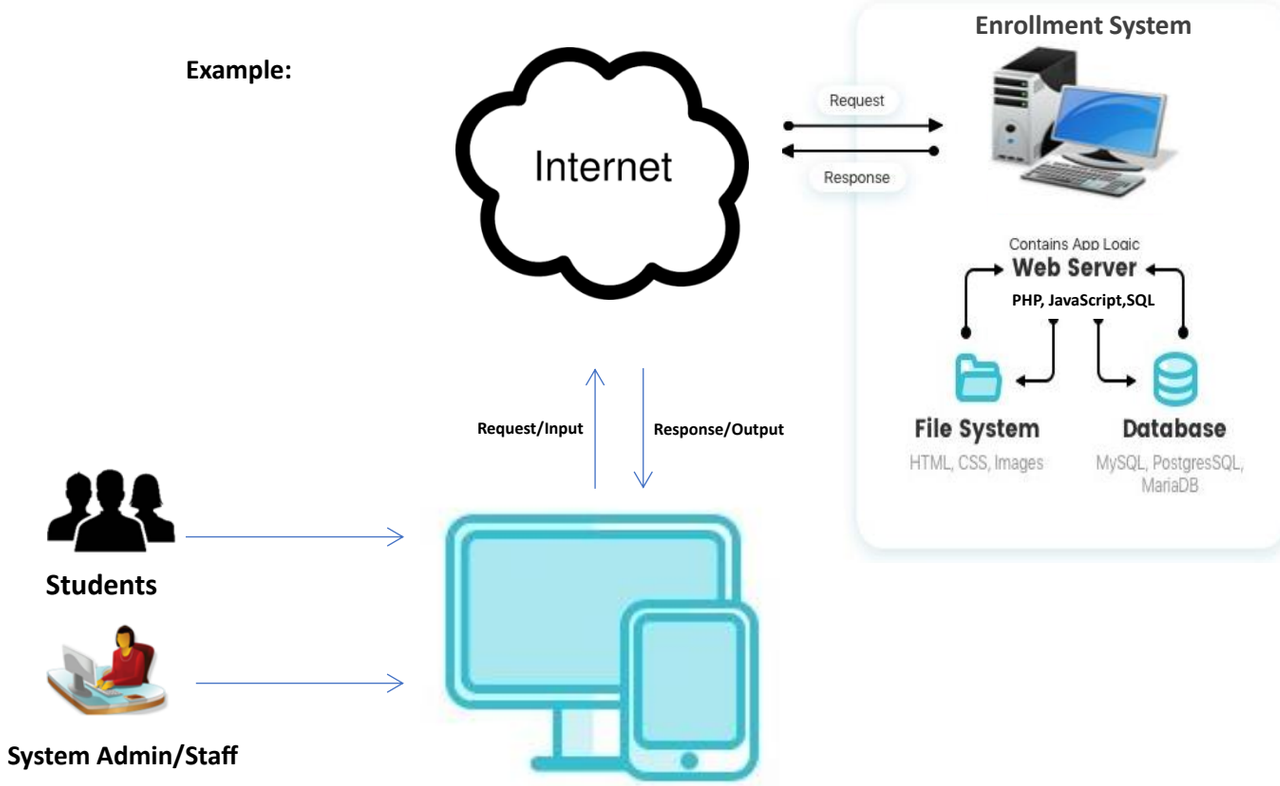
-External Entities (rectangle)
- Process (square with rounded corners with process number at top)
- Data Store
- Data Flow Lines (in and out)
- Main System (big square with rounded corners)

Sample:



- **System Architecture Design** - Provide a detailed illustration of the System Architecture Design, showcasing the overall structure of the system. The design should include key components such as user types, network devices, cloud services (if applicable), and the developed system. Highlight the relationships and interactions between these components to give a clear overview of how the system will function.
    - Support the illustration with a discussion by a team member, explaining each component and its role in the system. The discussion should emphasize how the architecture supports the system's objectives, scalability, and efficiency, providing a comprehensive understanding of the system's deployment and operation.

Example:



Important Note: The system architecture design is flexible and does not adhere to a strict format. You are encouraged to create your own design, provided it clearly and effectively illustrates the structure, components, and interactions within the system. Ensure that the design is logical, comprehensive, and aligns with the objectives of the project.

What to Prepare: You may prepare snacks (food and drinks) for 4 people. ***This is optional but recommended.***

Attire: Type A Uniform or Smart Casual

Defense Duration: Approximately 60 Minutes/ 1 Hour

Defense Fee: *Php250.00*

Project Adviser's Fee: *Php280.00*

Final Schedules of CP Teams: Will be posted soon in the official ITCP Class - FB Group

Prepared by:	Noted:
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